Finance & Habit Pattern Finder

Topic 9 – Google Opal Project

Personalized Transaction Analysis & Financial Advice

1. Project Overview

Finance & Habit Pattern Finder is an AI-powered financial transaction analysis workflow developed using Google Opal.

The project uses persistent memory to retrieve historical transactions, analyze current spending activity, identify patterns, and generate personalized financial recommendations.

It demonstrates how user inputs, AI reasoning, memory tools, and browser-based reporting can work together in an automated workflow.

2. Project Objectives

- Capture daily transaction details.
- Retrieve historical financial records from persistent memory.
- Compare current transactions with historical spending.
- Identify spending increases and repeated category activity.
- Generate personalized financial recommendations.
- Append new transaction records without overwriting previous history.
- Display financial insights through a browser-based report.

3. Key Features

- **Transaction Input:** Collects date, amount, category, and description.
- **Persistent Memory:** Maintains transaction history across workflow executions.
- **Spending Analysis:** Compares current transactions with available historical records.
- **Pattern Detection:** Identifies repeated spending and category trends.
- **Financial Recommendations:** Generates practical action plans based on recorded data.

- **Financial Dashboard:** Displays transaction summaries, weekly insights, and a complete available transaction archive.

4. Workflow Architecture

The project follows this workflow:

```
None
Transaction Date ─┐
Amount ───────────┤
Category ─────────┼──> Process Transaction Analysis
Description ──────┘                |
                                   v
                              Use Memory
                                   |
                                   v
                         Generate Financial Report
                                   |
                                   v
                              Browser Output
```

5. Technology and Tools

- Google Opal
- AI Generate nodes
- User Input nodes
- Use Memory tool
- Markdown documentation
- GitHub public repository

6. Repository Structure

```
None
finance-habit-pattern-finder/
|
├── prompts/
|   ├── transaction_analysis_prompt.md
```

```
|   └── financial_report_prompt.md
|
├── concept_guide.md
├── requirement_document.md
├── implementation_plan.md
├── testing_plan.md
└── README.md
```

7. Sample Transaction Analysis

The project was tested using three sample Food transactions:

Date	Description	Category	Amount
2026-11-06	Previous lunch	Food	$120
2026-11-07	Family dinner	Food	$450
2026-11-08	Restaurant lunch	Food	$180

Total recorded spending: $750

The final analysis identified a $60 difference and a 50% increase between the two similar lunch transactions.

8. Testing Summary

Test Case	Validation	Status
Test 1	Initial transaction handling	PASS
Test 2	Historical transaction comparison	PASS

Test 3	Repeated category activity and complete archive	PASS

See testing_plan.md for the detailed test scenarios, expected results, and observations.

9. Project Documentation

- Concept Guide
- Requirement Document
- Implementation Plan
- Testing Plan
- Transaction Analysis Prompt
- Financial Report Prompt

10. Scope and Limitations

This is an educational financial analysis workflow.

It does not connect directly to real bank accounts, process payments, or transfer money.

Analysis accuracy depends on the transaction details and historical records available to the workflow.

11. Conclusion

Finance & Habit Pattern Finder demonstrates how Google Opal can combine AI reasoning, persistent memory, transaction analysis, and browser-based reporting to support better financial spending awareness.

Project: Finance & Habit Pattern Finder

Topic: 9

Platform: Google Opal